

CASE STUDY / NATIONAL PLATFORM

The system running Pakistan's national gaming and animation program.

CEGA is a government-backed hub for training, startup incubation, co-working, and community events, used by more than 10,000 students. I designed and built the whole platform on my own, replacing an old WordPress site that could not keep up with the program it was meant to run.

10,000+

STUDENTS SERVED

3,392

APPLICANTS IN ONE BATCH

200+

STARTUPS INCUBATED

14+

APPLICATION PORTALS

MY ROLE

Full-stack developer, solo build and maintenance

CLIENT

CEGA, Center of Excellence in Gaming & Animation

BACKED BY

Ministry of IT & Telecom, Ignite National Technology Fund

BUILT WITH

Next.js, MongoDB, OpenAI, NextAuth

CENTERS

Karachi and Lahore, plus online

STATUS

Live at cega.com.pk

Read this in three layers: the story of the problem and the fix, the automation that now runs without staff, and a plain-English breakdown of the technology for both technical and non-technical readers.

THE PROBLEM

Growing attention, no system to catch it

CEGA runs Pakistan's national training program for gaming and animation. When I came in, the entire thing ran on a basic WordPress website.

It barely showed up in Google, so the growing interest the program was earning had nowhere useful to land. More importantly, there was no real system behind the site. There was no proper way to take and verify student applications, no way for judges to score applicants fairly, no automatic emails when someone applied or got accepted, and no way to book a co-working seat, a rendering lab, or a meeting room online. Everything happened by hand, or did not happen at all. Rising demand was a problem instead of an asset, because the site had no way to turn visitors into enrolled students, scored applicants, and booked sessions.

WHAT CEGA ACTUALLY IS

Four programs, one platform

CEGA is not a single website. It is four connected programs, each with its own applications, data, and reporting, all run from one system I built and maintain.

01 / TRAINING

Courses and enrollment

Applications, enrollment, and outcomes across multiple batches and courses like Unity 3D, Unreal Engine, and 3D animation, onsite in Karachi and Lahore and online.

02 / INCUBATION

Startup evaluation

A two-phase judging process that screens founders on market fit, then runs a deeper investment-grade review on the finalists.

03 / CO-WORKING

Seats and labs

Bookings and occupancy for co-working seats, meeting rooms, and a rendering lab, tracked per center.

04 / COMMUNITY

Events and participation

Event hosting, bookings, and participant tracking that ties the whole community together.

WHAT RUNS ON ITS OWN

The work the platform does without staff

This is the real value of the build. Jobs that used to need a person now happen automatically, correctly, every time.

01 Student applications and verification

Multi-step forms that check and save answers as the student types, then confirm the submission. No manual data entry.

Before: paper and inbox chaos

Now: self-serve and verified

02 Emails that send themselves

A new sign-up, a booking, a judge's score, or an accept or reject decision each trigger the right email on their own.

Before: someone types each reply

Now: sent automatically

03 Two-phase judge scoring

Applicants are scored in two rounds. Each judge sees only their own list, and the system stops any judge from overwriting another's score.

Before: spreadsheets, disputes

Now: tracked and locked

04 Bookings with conflict prevention

Seats, rooms, and the rendering lab are booked online, and the system blocks double-bookings before they happen.

Before: clashes and guesswork

Now: real-time availability

05 AI assistant, 24/7

A chatbot answers applicant questions about courses, programs, and locations instantly, at any hour, so the team stops repeating itself.

Before: same questions daily

Now: answered instantly

06 One live dashboard

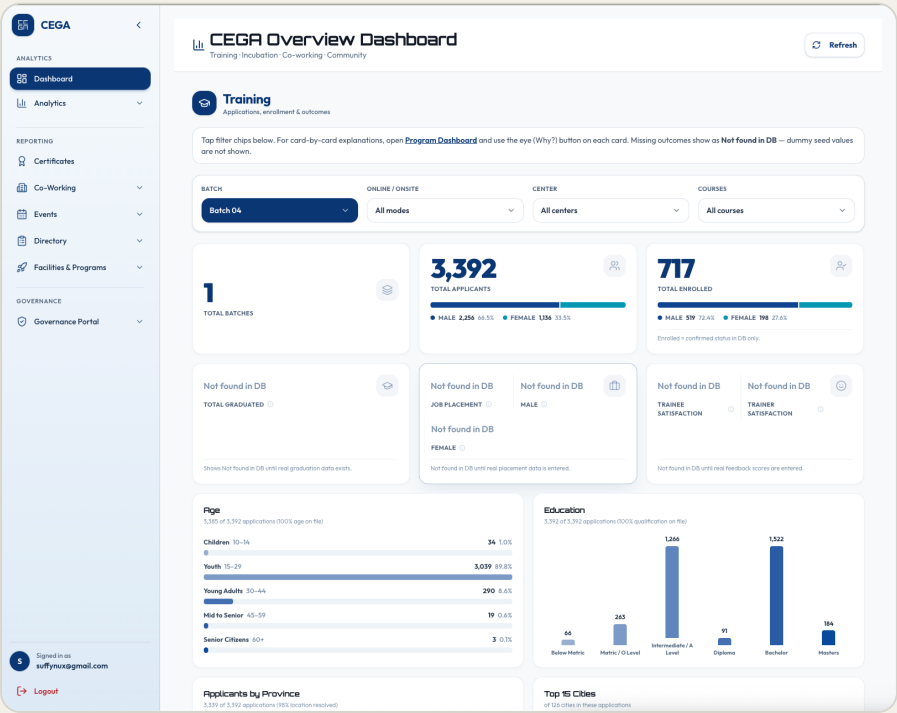
Applications, enrollment, bookings, and attendance roll up into one dashboard, filterable by batch, city, and date.

Before: scattered spreadsheets

Now: one live view

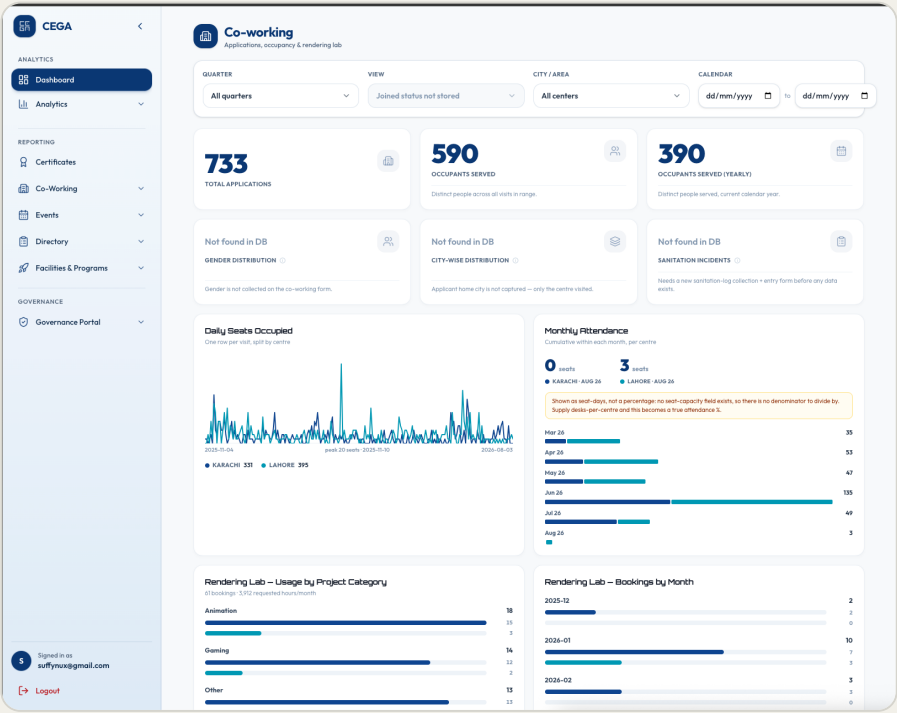
INSIDE THE PLATFORM

Real dashboards, real numbers



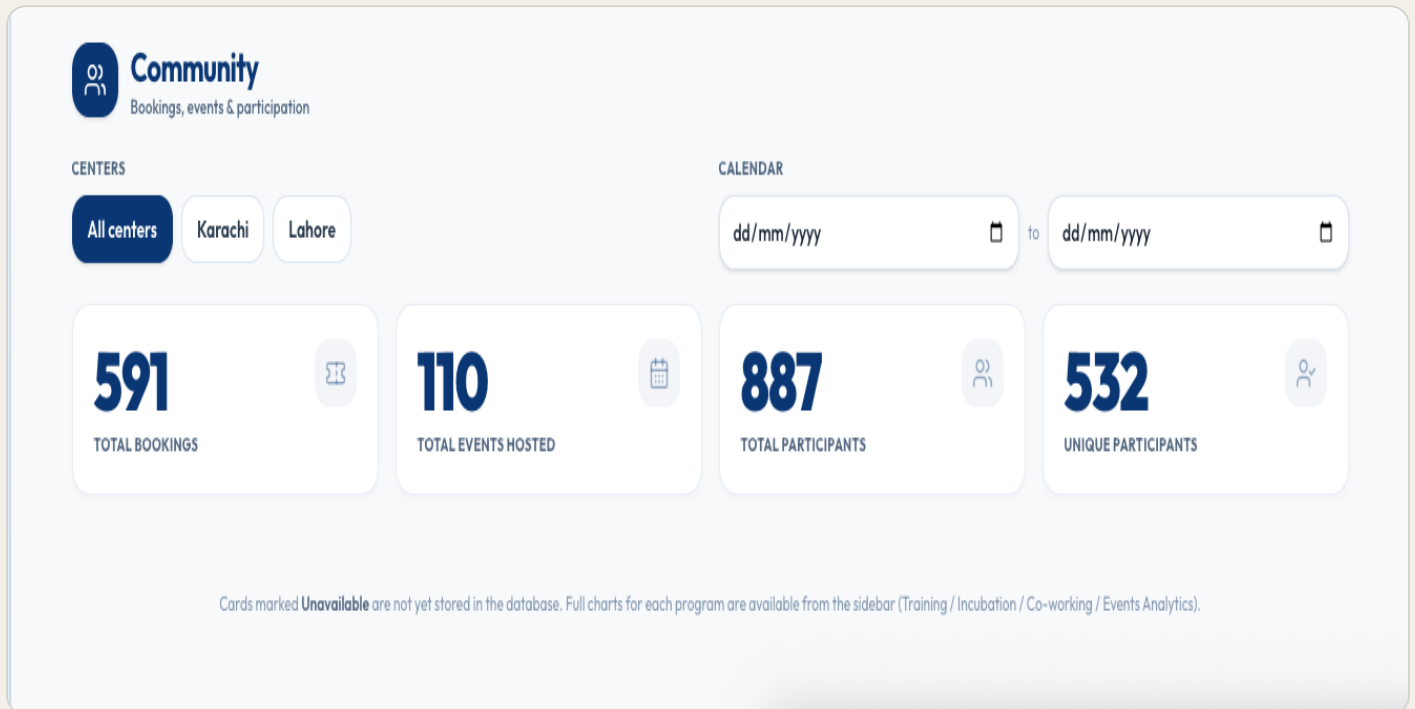
Training overview

3,392 applicants and 717 enrolled students in a single batch, broken down live by age, education, gender, and city across 126 cities.



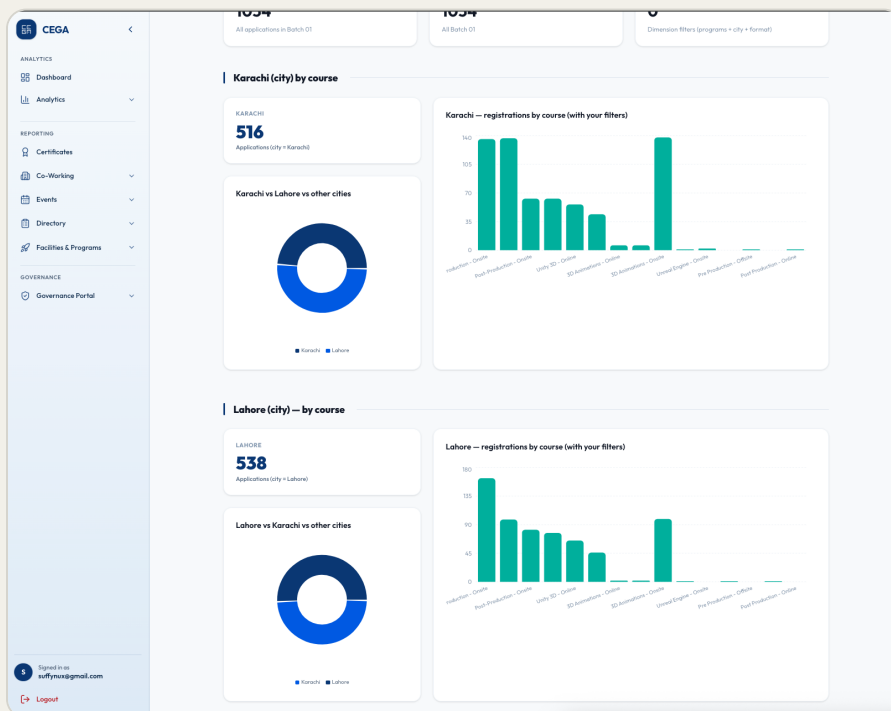
Co-working and rendering lab

733 applications and 590 people served, with daily seat occupancy per centre and 61 rendering-lab bookings tracked against requested hours.



Community and events

591 bookings, 110 events hosted, and 887 participants (532 unique), filterable by center and date range.



City-level breakdown

Registrations by course for Karachi (516) and Lahore (538), each with its own live filters for batch, delivery mode, and course.

FAIR, TRACKED JUDGING

Two rounds of scoring, no disputes

The screenshot shows the CEGA Phase 1: Concept & Market Fit judging interface. On the left, a sidebar lists applicants, with Haider Rasheed selected. The main panel displays the applicant's details, including their location (Lahore), team size (Game Tools / Sof...), and team lead (Haider Rasheed...). The 'CONTEXT' section shows a problem statement and a solution summary. The 'Executive Scores' section includes a table for judges to provide scores across six criteria: Sector Fit, Product Development Stage, Application Completion Status, Team Strength, Market Potential, and Program Fit. A 'JUDGE NOTES' section is also present for providing professional feedback. The interface includes a 'FINALIZE SCORES' button and a 'VIEW SCORING LEXICON' link.

Phase 1, concept and market fit

Each applicant is scored across six weighted criteria, with a composite average and a qualify or reject status.

The screenshot shows the CEGA Phase 2: Final Evaluation judging interface. On the left, a sidebar lists applicants, with Arif Majeed Khan selected. The main panel displays the applicant's details, including their location (Lahore), team size (Casual Game), and team lead (Arif Majeed, L...). The 'FINAL REASSESSMENT CONTEXT' section shows a problem statement and a solution summary. The 'Investment Grade Matrix' section includes a table for judges to provide scores across six criteria: Sector Fit, Product Development Stage, Application Completion Status, Team Strength, Market Potential, and Program Fit. A 'JUDGE NOTES' section is also present for providing professional feedback. The interface includes a 'FINALIZE SCORES' button and a 'VIEW SCORING LEXICON' link.

Phase 2, final evaluation

An investment-grade review on the finalist pool, with a portfolio-readiness score.

The screenshot shows the CEGA AI Assistant interface. It features a chat window with a blue header containing the CEGA logo and the text 'CEGA AI Assistant Powered by intelligent assistance'. The chat history shows a message from the user asking 'I want to know about the training module' and a response from the assistant stating 'CEGA offers comprehensive training in areas such as Unity 3D, Unreal Engine, 3D Animation, and Pre/Post Production, available in Lahore, Karachi, and online. Our programs also include entrepreneurship courses, digital skills training, and business development. For more details, feel free to browse our website at cega.com.pk!'. The interface includes a 'Ask anything...' input field and a send button.

AI assistant

Answers course questions in real time.

THE TECHNOLOGY

What it is built with, in plain words

Every part of the platform is listed below twice: once in plain English, and once with the actual tool. A non-technical reader can follow the middle column; a technical reader can check the right one.

PART OF THE SYSTEM	WHAT IT DOES	BUILT WITH
The website and pages	Everything visitors see: fast pages, forms, and dashboards that work on phone and desktop.	Next.js 15, TypeScript, Tailwind CSS
The database	Where every applicant, booking, score, and record is stored and searched.	MongoDB Atlas, Mongoose
Logins and roles	Secure sign-in that shows admins, judges, and students only what they are allowed to see.	NextAuth, JWT, bcrypt
The AI assistant	The chatbot that answers questions about courses and programs on its own.	OpenAI GPT-4
Automatic emails	Confirmations, booking notices, and status updates that send themselves.	NodeMailer
Images and files	Uploads and media handled and delivered quickly.	Cloudinary
Hosting	Keeps the site fast and online for thousands of users.	Vercel edge network

The hardest part was the judge scoring. Several judges can score the same applicant at the same time, each score is recorded and tied to that judge, and the system makes sure no one can ever overwrite another judge's work.

Next.js 15

TypeScript

MongoDB

NextAuth

OpenAI GPT-4

NodeMailer

Cloudinary

Tailwind CSS

Vercel

WHAT THIS MEANS FOR YOU

The same parts, any size project

CEGA is a large platform, but it is built from the same building blocks I use on smaller jobs. Whatever you need, one part of this project already proves I can do it.

WEBSITE

Need a website? This is the same foundation, just smaller and faster to build, and made to actually show up in search.

AI

Want AI on your site? The assistant here is the same kind of AI feature I can add to yours, for support, sales, or bookings.

AUTOMATION

Tired of manual work? The automatic emails, scoring, and bookings are automation. The same idea removes busywork like invoices, reminders, and reports.

WEB APP

Need something bigger? This handles logins, roles, dashboards, and thousands of users without breaking. Your custom tool or portal is well within range.

LET'S BUILD YOURS

Have a project in mind?

A website, an AI feature, an automation, or a full web app. One developer, start to finish.

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